

Note the quiz is front & back**SHOW YOUR WORK**Points: ____/16_{16.5}

1. Bonus. Describe what the ancient Romans used caltrops for, which have a tetrahedral geometry. (0.5 pts)

2. A4.104. Provide the correct IUPAC name for N₃O₅. (1 pt)

3. A5.2. How many valence electrons does bromine (Br) have? (1 pt)

4. ~ ICE. Which of the following is the correct Lewis symbol for the nitride ion? **Circle it & write in its charge.** (1 pt)



a. Draw/write the Lewis symbol for **carbon** using dots to show the required electrons. (1 pt)

5. A5.27. Draw the Lewis structure of O₂. (1.5 pts)

a. What is the total number of valence electrons in the above formula? (0.5 pts)

6. A5.114. Draw the Lewis structure for ammonium (NH₄⁺). (1 pt)

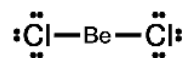
a. Identify the formal charge for all atoms that have non-zero formal charges. (0.5 pts)

7. ~ A5.78 & in class. Draw the Lewis structure of NO₂ and then calculate the appropriate formal charge for all atom(s) that have a non-zero formal charge. (1.5 pts)

a. Do all atoms in the above problem follow or are below the octet rule? **Yes or no** (0.5 pts)

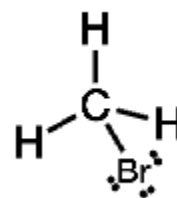
8. A5.125. Draw the Lewis structure of BH_3 and then determine its electron domain and molecular geometries. Note they *may* be the same. (2 pts)

9. The correct Lewis structure for BeCl_2 is given. Identify what the bond angle is around the central Be. (0.5 pts)



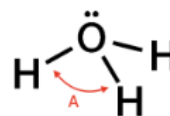
- a. How many sigma (σ) and pi (π) bonds are present in BeCl_2 ? Clearly identify numbers for both. (1 pt)

10. ~A5.158. The Lewis structure for bromomethane is provided. Does this molecule possess polar covalent bond(s)? (0.5 pts)



- a. Is this molecule polar or nonpolar? (1 pt)

11. A5.143. A Lewis structure for the hydronium ion (H_3O^+) is shown below on the left. Predict whether bonding angle A will be **equal to**, **greater than**, or **less than** the ideal bonding angle according to the VSEPR model. (1 pt)



- a. What is the ideal bond angle? (0.5 pts)

Period

Group

1

18

1 H 1.008 hydrogen																	2 He 4.003 helium						
2																							
3 Li 6.94 lithium	4 Be 9.012 beryllium															5 B 10.81 boron	6 C 12.01 carbon	7 N 14.01 nitrogen	8 O 16.00 oxygen	9 F 19.00 fluorine	10 Ne 20.18 neon		
11 Na 22.99 sodium	12 Mg 24.31 magnesium																	13 Al 26.98 aluminum	14 Si 28.09 silicon	15 P 30.97 phosphorus	16 S 32.06 sulfur	17 Cl 35.45 chlorine	18 Ar 39.95 argon
3																							
19 K 39.10 potassium	20 Ca 40.08 calcium	21 Sc 44.96 scandium	22 Ti 47.87 titanium	23 V 50.94 vanadium	24 Cr 52.00 chromium	25 Mn 54.94 manganese	26 Fe 55.85 iron	27 Co 58.93 cobalt	28 Ni 58.69 nickel	29 Cu 63.55 copper	30 Zn 65.38 zinc	31 Ga 69.72 gallium	32 Ge 72.63 germanium	33 As 74.92 arsenic	34 Se 78.97 selenium	35 Br 79.90 bromine	36 Kr 83.80 krypton						
4																							
37 Rb 85.47 rubidium	38 Sr 87.62 strontium	39 Y 88.91 yttrium	40 Zr 91.22 zirconium	41 Nb 92.91 niobium	42 Mo 95.95 molybdenum	43 Tc [97] technetium	44 Ru 101.1 ruthenium	45 Rh 102.9 rhodium	46 Pd 106.4 palladium	47 Ag 107.9 silver	48 Cd 112.4 cadmium	49 In 114.8 indium	50 Sn 118.7 tin	51 Sb 121.8 antimony	52 Te 127.6 tellurium	53 I 126.9 iodine	54 Xe 131.3 xenon						
5																							
55 Cs 132.9 cesium	56 Ba 137.3 barium	57-71 La-Lu *	72 Hf 178.5 hafnium	73 Ta 180.9 tantalum	74 W 183.8 tungsten	75 Re 186.2 rhenium	76 Os 190.2 osmium	77 Ir 192.2 iridium	78 Pt 195.1 platinum	79 Au 197.0 gold	80 Hg 200.6 mercury	81 Tl 204.4 thallium	82 Pb 207.2 lead	83 Bi 209.0 bismuth	84 Po [209] polonium	85 At [210] astatine	86 Rn [222] radon						
87 Fr [223] francium	88 Ra [226] radium	89-103 Ac-Lr **	104 Rf [267] rutherfordium	105 Db [270] dubnium	106 Sg [271] seaborgium	107 Bh [270] bohrium	108 Hs [277] hassium	109 Mt [276] meitnerium	110 Ds [281] darmstadtium	111 Rg [282] roentgenium	112 Cn [285] copernicium	113 Nh [285] nihonium	114 Fl [289] flerovium	115 Mc [288] moscovium	116 Lv [293] livermorium	117 Ts [294] tennessine	118 Og [294] oganesson						

Periodic Table of the Elements

Atomic number → 1

Symbol → **H**

Atomic mass → 1.008

Name → hydrogen

Color Code

	Metal		Solid
	Metalloid		Liquid
	Nonmetal		Gas